Cost & Architecture Report

Project Overview

This report summarizes the deployment and operational costs of the project hosted on Railway cloud infrastructure. The project uses Railway-managed compute and memory resources for application hosting and networking. Cost estimates are based on the actual usage data captured from the Railway dashboard.

1. Architecture Summary

The application is deployed on Railway using a containerized cloud environment. The architecture includes:

- Application Runtime Service hosted on Railway
- Memory allocation for application execution
- CPU resources for processing requests
- Network egress for outbound traffic
- Cloud-managed deployment and scaling

Railway automatically provisions and manages the infrastructure, reducing the need for manual server management while providing scalable deployment capabilities.

2. Itemized Cost Breakdown

Resource	Usage	Pricing	Estimated Cost
Memory	122.26 minutely GB	$0.000231 / GB / Minute	$0.0283
CPU	0.24 minutely vCPU	$0.000463 / vCPU / Minute	$0.0001
Network Egress	0.00 total GB	$0.05 / GB	$0.0000

Current usage cost recorded from the Railway dashboard: $0.03

Estimated projected usage cost: $0.04

3. Cost Dashboard Screenshots

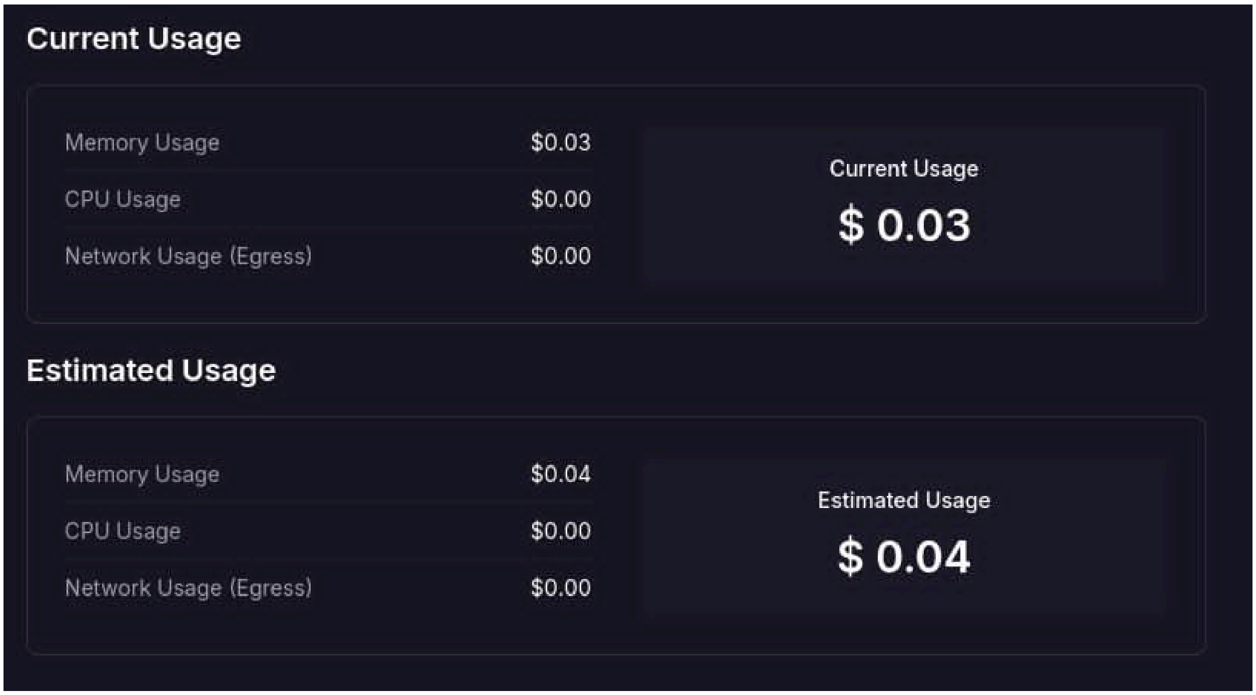

Figure 1: Railway Current and Estimated Usage Dashboard

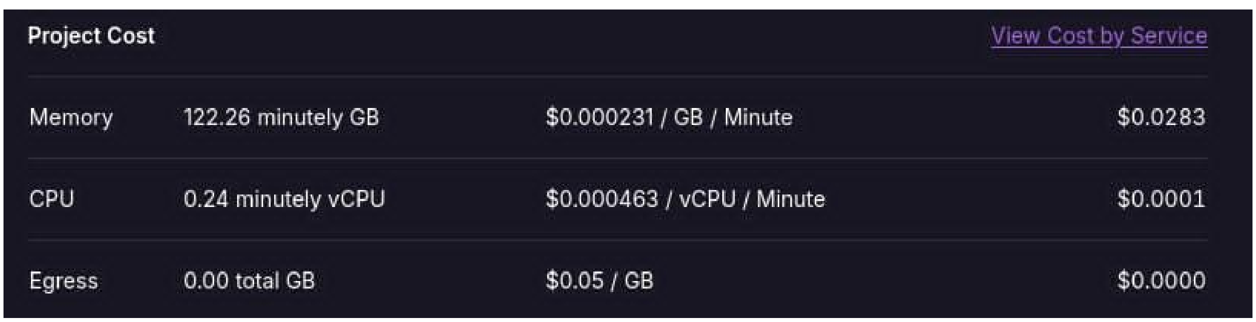

Figure 2: Railway Detailed Project Cost Breakdown

4. Cost Optimization Strategy

A realistic cost optimization strategy is to reduce allocated memory usage by optimizing application processes and enabling automatic idle shutdown for unused services.

Estimated Savings:

• Reducing memory consumption by 30% could reduce the memory cost from approximately $0.0283 to around $0.0198.

• Estimated monthly savings: approximately 25–30% of the total infrastructure cost.

• Additional savings can be achieved by minimizing unnecessary background services and optimizing application runtime efficiency

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5. Conclusion

The Railway deployment currently operates at a very low infrastructure cost, making it suitable for small-scale or academic projects. Memory usage is the primary contributor to the total cost, while CPU and network egress costs remain minimal. Applying optimization strategies can further improve cost efficiency without affecting application functionality.